

Deepak Mallampati

APPLIED AI / LLM ENGINEER

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US-based, F-1 OPT · open to relocation + EU sponsorship

PROFESSIONAL SUMMARY

Applied AI / LLM Engineer with 4+ years building production-grade LLM and agentic systems for banking and regulated enterprises. Specializes in the LLM application and orchestration layer - prompts, retrieval, tool calling, agents, and structured outputs - backed by rigorous evals (RAGAS, BERTScore), observability, and safe-rollout practices. Built RAG pipelines reaching **92% precision** with **40% hallucination reduction**, and a conductor-subagent orchestration framework that cut token cost by **42%** while holding accuracy. Pragmatic about model routing, semantic caching, and cost/latency trade-offs for enterprise-grade AI features.

CORE COMPETENCIES

LLM Orchestration • RAG & Vector Search • Agent Frameworks • Structured Outputs / Tool Calling • Evals (RAGAS, BERTScore) • MLOps & Observability • Model Routing & Caching • Anomaly Detection

Generative AI & LLMs: LangChain/LangGraph, RAG architecture, multi-agent systems, structured outputs, tool calling, prompt engineering, fine-tuning (LoRA/QLoRA)

Retrieval & Vector: FAISS, Pinecone, Weaviate; semantic & hybrid search; cross-encoder re-ranking; chunking; semantic caching

Evals & MLOps: RAGAS, BERTScore, custom metrics; drift detection; MLflow, W&B; model monitoring & observability; safe rollout

Backend & Infra: Python, FastAPI; Docker, Kubernetes; micro-services; Kafka (event-driven); PostgreSQL, MongoDB, Redis; AWS/Azure/GCP

Security & Compliance: HIPAA, SOC 2, PCI DSS; PII redaction; prompt-injection mitigation; audit logging

PROFESSIONAL EXPERIENCE

Progress Solutions

Jul 2025 - Present

AI Engineer

USA

- Architected production **agentic LLM systems** on a conductor-subagent orchestration framework, reducing token consumption by **42%** while sustaining accuracy across multi-step workflows.
- Engineered **RAG** pipelines over large document corpora combining dense retrieval with cross-encoder re-ranking, lifting answer relevance and reducing hallucination on domain queries.
- Established an **LLM evaluation & monitoring framework** (RAGAS, BERTScore, custom metrics) with latency/accuracy dashboards to catch regressions before release.
- Optimized inference cost/latency via prompt compression, **semantic caching, and model routing**, sustaining SLA targets.
- Built fault-tolerant automation with batch orchestration and exponential-backoff retries, reducing failed production runs by **60%**.

Vanguard

Jan 2024 - Jan 2025

AI Engineer Intern

USA

- Developed AI agents and backend services across **25+ AI agents** for Vanguard's internal AI platform, integrating production data pipelines and observability.
- Evaluated GPT-4o, Claude Sonnet, and Gemini; improved task success by **27%** and informed model-selection decisions.
- Built **12 APIs and microservices**, cutting p95 latency from 1.5s to **800ms** and supporting **100,000+ requests daily**.
- Engineered content controls for policy-compliant responses, reducing unsafe outputs by **42%**.

Kent State University

Oct 2023 - Jan 2024

ML & Gen AI Research Assistant

USA

- Built a privacy-preserving ML pipeline (k-anonymity, l-diversity, differential privacy), cutting re-identification risk from 3.38% to **0.32%** while retaining **99%** of baseline performance.
- Fine-tuned Qwen3-27B via 4-bit QLoRA on an H100 for controllable long-form generation; authored an IEEE-format paper.

Emerson

Jun 2022 - Apr 2023

ML Trainee

India

- Built regression/classification models for equipment-failure prediction and **anomaly detection** on operational data.
- Fine-tuned BERT/RoBERTa for entity extraction on financial datasets, achieving **89% F1** on custom NER.

PROJECTS

career-ops

Autonomous, eval-driven AI job-search pipeline that scans, scores, and tailors applications overnight without supervision - a zero-to-one shipped system.

MangaLens [Live on Chrome Web Store](#)

Real-time AI translation extension for manga/webtoons across 29+ sites, turning a manual chore into a single inline action.

EDUCATION

M.S. in Computer Science - Kent State University, OH

2025 · GPA 3.70/4.0